

Vikram Kamath Cannanure

POSTDOCTORAL FELLOW - SAARLAND UNIVERSITY · HCI, ICTD, AI, EDUCATION

✉ vica001@teams.uni-saarland.de · 🏠 kmarkiv.com · 🌐 vikramkamathc · 🎓 Google Scholar

I am a Postdoctoral Fellow at Saarland University (Ph.D., Carnegie Mellon University). I design Thoughtful AI—human-centered systems for reflection, care, and trust in the Global South. My work focuses on designing and deploying digital AI systems in real-world settings, grounded in **Human-Computer Interaction (HCI)** and international development.

- **Digital product development:** end-to-end AI systems (research → UX → engineering → deployment → evaluation)
- **Real-world deployments:** teacher chatbot in Ivory Coast(400+ teachers, 10,000+ students); Gates-funded health AI in rural India
- **Engineering & scale:** 10+ years (mobile, web, backend); deployments across Africa, South Asia, Europe
- **Research & leadership:** 18 full papers in top tier venues, raised over \$215,000; co-lead *HCI Across Borders*, mentored 25+ students

Research Experience

Saarland University

POSTDOCTORAL FELLOW

Saarbrücken, Germany

October 2023 – present

- Host: Prof. Ingmar Weber
- Leading research area at the intersection of HCI, ICTD, and Generative AI
- Co-supervising PhD and Master's students on socially grounded AI projects
- 🤝 Co-PI on a \$100,000 Gates Foundation grant to co-design a chatbot improving women's and youth health in rural India
- Instructor for the graduate seminar "AI and the Global South"
- Key publications: [P13, P12, P14, S4]

Education

Carnegie Mellon University

PhD, HUMAN-COMPUTER INTERACTION

Pittsburgh, USA

2017– Aug 2023

- Advised by Prof. Tim Brown and Prof. Amy Ogan
- Committee: Prof. Chinmay Kulkarni, Prof. Mayank Goel, and Prof. Kentaro Toyama (University of Michigan)
- Dissertation: [DLA: Supporting Teacher Professional Development in Low Infrastructure Settings](#)
- Led multi-year fieldwork and chatbot deployments for teachers in rural Côte d'Ivoire
- 🏆 Awarded \$100,000 Fellowship from the Center for Machine Learning and Health (CMLH)
- Key publications: [P6, P7, P8, P9, P10, P15]

Carnegie Mellon University

MASTERS, HUMAN-COMPUTER INTERACTION

Pittsburgh, USA

2017–2021

Carnegie Mellon University

MASTERS, EDUCATIONAL TECHNOLOGY AND APPLIED LEARNING SCIENCE

Pittsburgh, USA

2014–2015

National Institute of Technology Karnataka

BACHELOR OF TECHNOLOGY

Karnataka, India

2007–2011

Grants (\$215K USD) and Awards

2026–	Listen at Scale Program , Voice-first AI assistant enabling safe, private access to sexual, reproductive, and mental wellness with Gram Vaani, Sarvam AI, and EkStep Foundation 5,000 USD	India
2025–	Gates Foundation, Grand Challenges India (Co-PI) , A LLM enabled community centered platform for sexual and mental wellness among youth and women in rural India (in progress) - 100,000 USD	India
2019	PhD Fellowship , Center for Machine Learning and Health - 100,000 USD	Pittsburgh USA
2025	SIGCHI Development grant , HCIXB Workshop at CHI 2023, 2025- 10000 USD	
2019	Best Paper Award , ACM COMPASS Paper	Accra, Ghana
2019	Poster Spotlight , ACM COMPASS Poster	Accra, Ghana

Selected Publications

Over 35 peer reviewed work, including 18 full papers in top-tier venues in HCI, including *CHI*, *ICTD*, and *COMPASS*.

As of 2026: **390 citations**, **h-index: 10** Google Scholar: [HVuuUzwAAAAJ](#) ORCID: [0000-0002-0944-7074](#)

[P18] **Cannanure, V.K.**, Yinkfu, B., Bryan, D., Amin, M., Weber, I., *Teacher Professional Development on WhatsApp with LLMs: Early Lessons from Cameroon*. Accepted at Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2026), Springer LNCS.

[P17] Wolf, S.*, Ogan, A.*, Jasińska, K.*, Kembou, S.*, **Cannanure, V.K.**, Whitehead, H., Corbett, J., Tanoh, F., Wortsman, B., Lim, J., Zhao, H., Kumar, A. *Impacts of Targeted Instruction with Mobile-based Support on Teachers and Students in Rural Côte d'Ivoire*. Accepted at American Educational Research Journal.

- [P16] Behmanush, H., Akhtari, F., Weber, I., **Cannanure, V.K.**, *Designing Safe and Accountable GenAI as a Learning Companion with Women Banned from Formal Education*. Accepted at Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT 2026), ACM.
- [P15] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T.X., Ogan A. *A Design-Based Research Approach to Bridge Teacher Aspirations and Goal-Setting*. *ACM Journal on Computing and Sustainable Societies*. [\[Link\]](#)
- [P14] Zhu J., Kempermann M., **Cannanure V.K.**, Hartland A., Navarrete R., Carteny G., Braun D., Weber I. *Learn, Explore and Reflect by Chatting: Understanding the Value of an LLM-Based Voting Advice Application*. *ACM CUI 2025*, Waterloo, Canada. [\[Link\]](#)
- [P13] Adhikari D.M., Hartland A., Weber I., **Cannanure V.K.**. *Exploring LLMs for Automated Generation and Adaptation of Questionnaires*. *ACM CUI 2025*, Waterloo, Canada. [\[Link\]](#)
- [P12] Behmanush H., Akhtari F., Nooripoor R., Weber I., **Cannanure V.K.** *Online Learning and GenAI: Supporting Women's Aspirations Amid Socio-Political Instability in Afghanistan*. *ACM COMPASS 2025* [\[Link\]](#)
- [P11] Verma R., Weber I., **Cannanure V.K.**. *Exploring AI in Education: Child-Computer Interaction in the Global South*. *ACM Journal on Computing and Sustainable Societies*. [\[Link\]](#)
- [P10] **Cannanure V.K.**, Ngoon T., Wolf S., Jasińska K., Brown T., Ogan A. *Understanding the Longitudinal Impacts of a Chatbot to Facilitate a Virtual Community of Practice for Teachers in Rural Côte d'Ivoire*. *ACM COMPASS 2024*, Delhi, India. [\[Link\]](#)
- [P9] Ngoon T., **Cannanure V.K.**, Wolf S., Jasińska K., Brown T., Ogan A. *"I believe I did not preach into the desert": Opportunities & Challenges in Scaling Teacher Mentorship through Mobile Technology in Rural Côte d'Ivoire*. *L@Scale 2024*, Atlanta, USA. [\[Link\]](#)
- [P8] **Cannanure V.K.**, Ávila E., Ngoon T., Adjii Y., Wolf S., Jasińska K., Brown T., Ogan A. *"We dream of climbing the ladder, for getting there, we have to do our job better": Designing for Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2022*, Seattle, USA. [\[Link\]](#)
- [P7] **Cannanure V.K.**, Brown T., Ogan A. *DIA: A Human-AI Hybrid Chatbot for Low-Resource Contexts*. *ICTD 2020*, Guayaquil, Ecuador. [\[Link\]](#)
- [P6] **Cannanure V.K.**, Souvenir J., Adjii Y., Apke H., Jasińska K., Brown T., Ogan A. *"I'm fine where I am, but I want to do more": Exploring Teacher Aspirations in Rural Côte d'Ivoire*. *ACM COMPASS 2020*, Guayaquil, Ecuador. [\[Link\]](#)
- [P5] Madaio M., Yarzebinski E., **Cannanure V.K.**, Zinszer B., Seri A., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *Collective Support and Independent Learning with a Voice-Based Literacy Technology in Rural Communities*. *CHI 2020*, Hawaii, USA. [\[Link\]](#)
- [P4]  Madaio M., **Cannanure V.K.**, Yarzebinski E., Zasacky S., Tanoh F., Hannon-Cropp J., Cassell J., Jasińska K., Ogan A. *"You Give a Little of Yourself": Family Support for Children's Use of an IVR Literacy System*. *ACM COMPASS 2019*, Accra, Ghana. (Best Paper Award) [\[Link\]](#)
- [P3] **Cannanure V.K.**, Brown T. *RASP-IVR: A Low-Cost Interactive Voice Response System*. *AfriCHI 2018*, Windhoek, Namibia. [\[Link\]](#)
- [P2] Onafuwa D., Early K., Vyas N., **Cannanure V.K.**, Mankoff J. *Understanding the Information Needs of Prospective Tenants*. *ACM COMPASS 2018*, San Francisco, USA. [\[Link\]](#)
- [P1] Bhat K*, Thinayane M*, Goldkind L., **Cannanure V.K.** *Critical Participatory Design: Reflections on Engagement and Empowerment in a Case of a Community-Based Organization*. *Participatory Design Conference (PDC 2018)*, Hasselt and Genk, Belgium. [\[Link\]](#)

Book chapters

- [B1] Gautam A., Wong-Villacrés M., Gaytán Lugo L.S., **Cannanure V.K.**, Vannini S., Jimenez A., Tuli A., Reynolds-Cuéllar P., Pei L., Alvarado Garcia A., Badillo-Urquiola K., Prietch S.S., Densmore M., Banganayi T., Dray S. *Human-Computer Interaction Across Borders: Perspectives from the Global South*. In *The Human-Computer Interaction Handbook* (4th ed.), Julie A. Jacko (Ed.). CRC Press, 2026.

Selected Short Papers

- [S5] **Cannanure V.K.**, Wolf S., Jasinska K., Brown T., Ogan A. *Enabling a Virtual Community of Practice Facilitated by a Chatbot for Rural Teachers. Proceedings of the 5th African Human-Computer Interaction Conference (AfriCHI 2025), Cairo, Egypt.* [\[Link\]](#)
- [S4] Mensah E.E., **Cannanure V.K.**, Weber I. *Ethnic Diversity and Spatial Dynamics in Google Maps Reviews: Insights from a German Border City. ICWSM 2025 (poster).* [\[Link\]](#)
- [S3] **Cannanure V.K.**, Wolf S., Jasińska K., Brown T.X., Ogan A. *Applying Think-Aloud in ICTD: A Case Study of a Chatbot Use by Teachers in Rural Côte d'Ivoire. ICTD 2024, Nairobi, Kenya.* [\[arXiv\]](#)
- [S2] Adhikari D.M., **Cannanure V.K.**, Hartland A., Weber I. *Exploring LLMs for Automated Pre-Testing of Cross-Cultural Surveys. ICTD 2024, Nairobi, Kenya.* [\[arXiv\]](#)
- [S1] **Cannanure V.K.**, Chen X., Mankoff J. *Twist and Knock: A One-Handed Gesture for Smart Watches. Proceedings of the 42nd Graphics Interface Conference (GI 2016), Victoria, Canada.* [\[Link\]](#)

Selected Workshops and Lightly reviewed work

- [L4] **Cannanure V.K.**, Varghese D., Rivera-Loaiza C., Noor F., Das D., Jain P., Chang-Rizzo M., Wong-Villacres M., Karusala N., Ahmed N., Till S., Akhigbe B., Dray S., Densmore M., Sturm C., Kumar N. (2023). *HCI Across Borders: Towards Global Solidarity.* CHI '23, Apr 2022, Hamburg, Germany.
- [L3] **Cannanure V.K.**, Karusala N., Rivera-Loaiza C., Prabhakar A. S., Varanasi R. A., Tuli A., Gamage D., Noor F., Nemer, D., Das D., Dray S., Sturm C., Kumar, N. (2022). *HCI Across Borders: Navigating Shifting Borders at CHI.* CHI '22, Apr 2022, New Orleans, USA.
- [L2] **Cannanure V.K.**, Gamage D., Sturm C., Winschiers-Theophilus H., Maestre, J. F., Karusala N., Reynolds-Cuéllar P., Kumar, N., *Workshop: Decolonizing HCI Across Borders.* CHI '21, Apr 2021, Yokohama, Japan.
- [L1] Kumar, N, **Cannanure V.K.**, Gamage D. , Prabhakar A., Sturm C. *Workshop: HCI Across Borders and Sustainable Development Goals.* CHI '20, Apr 2020, Hawaii, USA

Teaching Experience

AI and the Global South (Block Seminar)

LEAD INSTRUCTOR (COURSE CREATOR)

Summer 2025

Saarland University

Programming Usable Interfaces (05-430/05630)

TEACHING ASSISTANT [HYBRID]. INSTRUCTOR: SCOTT HUDSON

Spring 2022

Carnegie Mellon University

User-Centered Research & Evaluation (05-410/05-610)

TEACHING ASSISTANT [REMOTE]. INSTRUCTOR: RAELEN MUSURACA

Spring 2021

Carnegie Mellon University

Elements of Data Science and Artificial Intelligence

COURSE COORDINATOR. INSTRUCTOR: INGMAR WEBER

Fall 2024

Saarland University

Professional Service

2026	Paper chair , ACM COMPASS 2026	Virtual
2025	Lead Mentor , HCI Across Borders (HCIXB) workshop at CHI 2025	Yokohama, Japan
2024	Lead Organizer , HCI Across Borders (HCIXB) workshop at ACM COMPASS	Delhi, India
2023	Lead Organizer , HCI Across Borders (HCIXB) workshop at CHI	Hamburg, Germany
2022	Lead Organizer , HCI Across Borders (HCIXB) workshop at CHI	New Orleans, USA
2021	Lead Organizer , HCI Across Borders (HCIXB) workshop at CHI	Yokohama, Japan
2020	Organizer , HCI Across Borders (HCIXB) symposium at CHI	Hawaii, USA
2019	Organizer , HCI Across Borders (HCIXB) symposium at CHI	Glasgow, UK
2019	Organizer , Young Researchers reception ICTDx	Ahmedabad
2020-22	SIGCHI Development Fund , Executive Committee	
2024	Programme Committee , COMPASS 2025, AFRICHI 2025, ICTD 2024	
2020-	AC , CHI (extended abstracts) 2023,2022,2021,2020	
2018-	Regular Reviewer (over 80 papers reviewed) , CHI, COMPASS, ICTD, CSCW, CHI case studies	

Mentoring

PhD

- ***Hamayoon Behmanush**: Saarland University, PhD: *Supporting Afghan women learn programming* (P12, P16)
- ***Ethel Mensah**: Saarland University, PhD: *Exploring structural inequality in Germany using POI data* (S4)

Masters

- ***Divya Mani Adhikari**: Saarland University, MS 2025: *Automated Survey Testing using LLMs* (P13, S2)
- Prithika Acharya**: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Ivory Coast*
- Aastha Patel**: CMU Pittsburgh, METALS 2022: *Teacher Training Project in Ivory Coast*
- Egide Kamuzinzi**: CMU Africa, MSIT 2021: *Teacher Training Project in Ivory Coast*
- Baptiste Moreau-Pernet**: CMU Pittsburgh, METALS 2021: *Teacher Training Project in Ivory Coast*
- Saumya Sharma**: CMU Pittsburgh, MSIPM 2020: *Simple Health Information System*
- Lucie Uwimpaye**: CMU Africa, MSIT 2021: *Simple Health Information System*
- Eloisa Ávila**: CMU Pittsburgh, MSPPM 2021: *Teacher Training Project in Ivory Coast*
- Sheila Mbadi**: CMU Africa, MSIT 2021: *Simple Health Information System*
- Patrick Manirakiza**: CMU Africa, MSECE 2019: *MnE system for Rwanda Women's Network*
- Solange Iyubu**: CMU Africa, MSECE 2019: *MnE system for Rwanda Women's Network*
- Sabin Sheja**: CMU Africa, MSECE 2019: *MnE system for Rwanda Women's Network*
- Justin Bigwi**: CMU Africa, MSECE 2019: *Design of a Phone Call-Based Health Information System*
- Gloria Umutesi**: CMU Africa, MSECE 2020: *Design of a Phone Call-Based Health Information System*
- Jocelyne Mukamisha**: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*
- Justin Souvenir Niweteto**: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System* (P6)
- Fraterne Kagame**: CMU Africa, MSIT 2020: *Design of a Phone Call-Based Health Information System*
- Huguette Ingenere**: CMU Africa, MSIT 2020: *French Interview Transcription for teachers in Cote d'Ivoire*
- Marie Claudine Ingenere**: CMU Africa, MSIT 2020: *French Interview Transcription for teachers in Cote d'Ivoire*
- Nebiyu Yismaw**: CMU Africa, MSECE 2018: *Digital and Voice based Systems to support GBV*
- Fabiola Uwashema**: CMU Africa, MSIT 2018: *Digital and Voice based Systems to support GBV*

Undergraduate

- ***Riti Verma**: Indian Institute of Technology, Delhi: *Exploring the role of AI in the Global South* (P11)
- Lena Armstrong**: University of Pennsylvania, Summer REU: *Teacher Training Project in Ivory Coast*
- Maya Greenholt**: CMU Pittsburgh, Cognitive Science: *Teacher Training Project in Ivory Coast*

*indicates during my postdoc at Saarland University

Professional Experience

Carnegie Mellon University, Africa

VISITING RESEARCHER

- Host: Prof. Tim Brown
- Project: Design of a Phone Call-Based Health Information System
- Conducted participatory research with rural women and prototyped an IVR system in rural Rwanda

Kigali, Rwanda
Summer - 2018, 2019, Spring 2020

United Nations University Computing and Society

RESEARCH INTERN

- Advisor: Dr. Mamello Thinayne
- Conducted participatory design with a local NGO
- Built a mobile app to report and monitor homelessness in Macau

Macau, SAR, China
July 2017 – Sept 2017

Carnegie Mellon University

SENIOR WEB DEVELOPER

- Advisor: Prof. Geoff Gordon
- Built and managed educational technology to support online learning
- Prototyping low cost video recording system for teaching courses

Pittsburgh, USA
July 2015 – June 2017

Carnegie Mellon University

RESEARCH ASSISTANT

- Advisor: Prof. Jen Mankoff
- Longitudinal eating detection: deployed and evaluated the system to 12 users in the field
- Twist n Knock: A one-handed Gesture for smart watches: publication at GI 2016
- EDigs: Built and maintained the APIs for apartment hunting website in Pittsburgh

Pittsburgh, USA
Aug. 2015 - June. 2017

TeliportMe Inc: Indian startup

WEB ARCHITECT

- Engineered APIs for Android, iOS, Web and Google Glass
- Scaled back-end to handle 4 million users and manage 100K locations
- Designed 3D viewer on par with Google street view
- Managed 10 Ubuntu machines with MySQL and Redis replication

Karnataka, IN
Jul 2011 - Aug 2014

Selected Press

- The Next Web** TeliportMe introduces 3D effects to its 360-degree panoramic app for Android
- Techcrunch** Test Flying: Applover Launches Free, Crowdsourced Testing Platform for Android Apps
- Robert Scoble** Developers: crowd-source your testing of Android phones
- CMU** A new online resource from Carnegie Mellon University seeks to help people find the perfect rental
- HCII** EDigs Helps Apartment Hunters Find the Perfect Rental

Skills

Programming Python, PHP, HTML/CSS, JavaScript, Android, Arduino, iOS
Technologies Pandas, Scikit, Weka, NumPy, AngularJS, Three.js, Bootstrap, MySQL, MongoDB, Redis, WebSockets
Coursework Applied Fabrication, Applied Machine Learning, Data Pipeline, Gadgets Activity and Recognition in HCI, Education Goals, E-Learning Design Principles, Game Design Studio, Design of Educational Games

Other

- **Philanthropy** Raised over \$20,000 for charity for causes in India through swimming and crowd-sourcing
- **Android** I have deployed 3 apps to the app store and have around 10,000 active users
- **Web** I have deployed over 20 websites to production. My best project has reached over 4 million users.
- **Teaching** I taught children a second language (Kannada) at the S.V. Temple in Pittsburgh
- **Swimming** I swam 10 miles from Elephanta Caves to Gateway of India to raise money for acid attack victims